Armin Salmasi

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LinkedIn   Google Scholar

ABOUT

Ph.D.-level R&D Engineer transitioning into Software Development and Machine Learning. I possess strong expertise in scientific computing, algorithm design, and building scalable data pipelines. My experience at Thermo-Calc Software involves full-cycle software development—from architecting physics-based prediction models (Python/Fortran) to deploying Java-based UIs and managing CI/CD pipelines. I have successfully engineered GenAI solutions using RAG and Large Language Models to automate code generation. With a deep background in High-Performance Computing (HPC) and data-driven modeling, I bridge the gap between complex domain logic and robust, production-ready software solutions.

TECHNICAL SKILLS

Languages	Python, Java, Fortran, CUDA, SQL, Bash Script, Batch Script
ML & AI	PyTorch, TensorFlow, Scikit-learn, LLMs (RAG, Prompt Eng.), LangChain, HuggingFace, MLFlow
Data Eng.	MongoDB, MySQL, SQLite, ETL Pipelines, Pandas, NumPy
DevOps	Docker, Git, Jenkins, Maven, Gradle, SVN, SLURM, Archiva
HPC	MPI, OpenMP, mpi4py, Distributed Computing
Sci. Comp.	Thermo-Calc, VASP, LAMMPS, COMSOL, ParaView, VTK
Tools	VSCode, IntelliJ, PyCharm, Jupyter, Linux, macOS, Windows

WORK EXPERIENCE

Software Engineer & Model Developer, Thermo-Calc Software AB *2023 - Present*

- Designed and implemented physics-based algorithms and data structures for the core calculation engine (Fortran).
- Developed and maintained the application's Graphical User Interface (GUI) using Java Swing and MigLayout.
- Engineered a Generative AI feature using RAG (Retrieval-Augmented Generation) to autonomously generate TC-Python scripts for users.
- Built and optimized data pipelines for extracting and assessing large-scale thermodynamic datasets.
- Managed the CI/CD pipeline, maintaining build environments using Jenkins, Docker, and Maven/Gradle.

Key Projects:

- **GenAI Integration:** Implemented LLM-based code assistants to improve user workflow efficiency.
- **Predictive Modeling:** Developed Python models for predicting yield strength and phase transitions in steel.
- **Database Development:** Structured and populated SQL/NoSQL databases for material property storage.
- **Legacy Refactoring:** Modernized legacy Fortran codebases and wrapped them for Python interoperability.

R&D Simulation Specialist, Epiroc Drilling Tools AB *2023*

- Utilized computational modeling to optimize production parameters, reducing trial-and-error costs.
- Analyzed large datasets from production testing to identify correlations in material performance.

Key Projects:

- Simulation of phase transformation dynamics under high-pressure conditions.
- Data-driven design optimization of cemented carbide grades.

Research Intern (Master Thesis), Sandvik Coromant AB *2016*

- Applied Density Functional Theory (DFT) calculations to screen high-entropy alloy candidates.

- Developed diffusion simulation models to predict gradient formation during sintering.

EDUCATION

Ph.D., Engineering Materials Science *2016 – 2022*
KTH Royal Institute of Technology
Focus: ICME, Mass Transport Simulation, and Numerical Modeling.

M.Sc., Materials Science *2016*
KTH Royal Institute of Technology
Focus: Simulation of Gradient Formation using Thermodynamic modeling.

M.Sc., Nano Materials *2011*
Materials & Energy Research Center
Focus: Thin Film Deposition and Characterization.

B.S., Extractive Metallurgy *2007*
Amirkabir University of Technology

CERTIFICATIONS & TRAINING

- **Supervised ML: Regression & Classification**, DeepLearning.AI 2023
- **Good Manufacturing Practice (GMP)**, Svensk Medicin AB 2022
- **Software Tools: ML to Phase Diagrams**, MGF 2022
- **Practical Deep Learning**, ENCCS 2021
- **CodeRefinery (Software Development)**, NeIC 2020
- **Structuring Machine Learning Projects**, DeepLearning.AI 2020
- **Improving Deep Neural Networks**, DeepLearning.AI 2020
- **Neural Networks & Deep Learning**, DeepLearning.AI 2020
- **Python: Functions & Dictionaries**, U. Michigan 2019
- **Python Basics**, U. Michigan 2019
- **The Unix Workbench**, Johns Hopkins University 2019
- **CAMD Summer School (DFT)**, DTU Physics 2018
- **CSC HPC Summer School**, Solvalla, Finland 2017
- **PDC HPC Summer School**, KTH, Sweden 2017
- **RACIRI Summer School on Materials Design**, Rugen 2015

SELECTED PUBLICATIONS

DIVA Portal — Google Scholar

1. **Houser et al.** A computational framework for modeling and predicting tensile behavior of materials applied to martensitic steels. *TMS, High performance steels. 2026*
2. **Salmasi et al.** Mobilities of Ti and Fe in Disordered TiFe-BCC Assessed from New Experimental Data. *CALPHAD. 2021*
3. **Salmasi et al.** Viscosity of Liquid Cobalt from Ab-initio Molecular Dynamics. *Manuscript.*
4. **Salmasi.** ICME Guided Study of Mass Transport in Production and Application of Cemented Carbides. *PhD Thesis. 2022*

RELEVANT COURSEWORK

- Advanced Computational Techniques
- Applied Numerical Methods
- Simulation and Modeling Toolbox
- First-Principles Tools (DFT)
- Diffusion in Multicomponent Solids
- Leadership & HR Management

TEACHING

T.A., Thermodynamics and Kinetics, KTH 2016-2021
T.A., Simulation & Modeling Courses, KTH 2017-2018
T.A., Advanced Course in Material Design, KTH 2018